

M.Loutfi Abouzaid

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Senior Advisor - Automation Software Engineer

EXPERIENCE

Preton, TX-USA- Remote

Python Automation Engineer

Nov 2024 - Present

Roles and Responsibilities:

- Develop and maintain automation frameworks to validate data integrity between Snowflake databases, Excel files, and UI elements.
- Design and execute Quality Improvement (QI) pipelines that process and validate millions of records across multiple relational tables.
- Build robust APIs using FastAPI to retrieve and process data from Snowflake and PostgreSQL, applying business logic to return actionable insights.
- Implement end-to-end automation scripts using Python, Pandas, and NumPy for data verification, transformation, and reporting.
- Optimize SQL queries and work with SQLAlchemy to ensure high performance and reliability in database interactions.
- Collaborate with QA, data, and engineering teams to ensure alignment of data models and accuracy across multiple systems.
- Troubleshoot automation issues, enhance test coverage, and document all validation logic to support long-term maintainability and scalability.

Ford Motor Company, FL-USA- Remote

Software Development Engineer in Test Feb 2023 - Sep 2024

Roles and Responsibilities:

- Developed automation frameworks and scripts using Python, Selenium WebDriver, and Pytest to automate validation sanity, regression, and integration test cases for both UI and RESTful APIs.

SKILLS

Python, Pytest, Selenium, API Testing, Backend, Building Frameworks, Data Analyst, Scrum, Agile, Automation scripting, Problem Solver, ETL, DBT, GCP AWS, CI/CD, MSSQL, Active Directory, BDD, Behave, Requests, Gherkin, Pandas, NumPy, Putty, PowerShell, Git, GitHub, Jenkins, JSON.

LANGUAGES

English

Arabic

- Implemented automated test sites that significantly improved testing efficiency and reduced manual efforts.
- Conducted OTA (Over-The-Air) testing to ensure the reliability and performance of wireless communication protocols, collaborating with the development team to identify and resolve issues related to OTA firmware updates.
Documented test plans, procedures, and results, ensuring comprehensive test coverage and utilizing industry-standard tools and equipment for OTA testing.
- Cleansed and manipulated database records using Pandas, NumPy, and Tableau for visualization, while integrating FastAPI, REST APIs, and JSON parsing for seamless data interactions.
- Designed and implemented ETL pipelines using Python to extract data from multiple sources, transform it into a unified format, and load it into Microsoft SQL Server databases, ensuring data integrity and consistency.
- Developed automated data extraction processes with Python, leveraging libraries such as pandas and SQLAlchemy, to streamline data ingestion into MS SQL Server for real-time analytics and reporting.
- Integrated error handling and logging mechanisms within Python-based ETL workflows, enabling robust monitoring and troubleshooting of data transfer processes between diverse data sources and MS SQL Server.
- Developed automated test scripts in Python to verify GCP services, APIs, and application functionality, focusing on accuracy and efficiency.
- Designed data validation tests using SQL and PostgreSQL validation techniques to ensure the integrity and accuracy of data stored in GCP databases.
- Conducted software testing and successfully delivered built solutions for vehicles, ensuring quality and reliability.
- Utilized Selenium WebDriver to automate UI interactions, ensuring accurate and reliable testing of web applications.
- Designed and executed comprehensive test plans, test cases, and test scripts for various functional and non-functional requirements.
- Collaborated with cross-functional teams, including developers, product managers, and QA engineers, to identify and address software defects and performance issues.
- Integrated automated tests into the CI/CD pipeline using GitLab Actions, enabling continuous testing and deployment.

- Implemented Pytest and Slash frameworks to test and validate functions built within the software, ensuring robustness and reliability.
- Obtained and evaluated information on factors such as reporting formats required, costs, and security needed to determine hardware configuration.
- Conducted code reviews, identified areas of improvement, and recommended best practices for code quality and testability.
- Deployed new fixes using SIT and Jenkins after a build is kicked off.
- Worked with JSON data to effectively test Backend/Database Testing.
- Executed Test Cases on Jira and TestRail on Benches and Vehicles.
- Wrote automation scripts in Python to run Sanity and Regression tests.
- Collaborated with engineers to develop effective strategies and test plans.
- Monitored known system errors, discussing resolution options with colleagues before implementation.
- Worked with cross-functional teams to provide optimal system functionality throughout the software development lifecycle.
- Created modular and extensible test frameworks using Python testing libraries, such as Pytest.
- Dealt with Automotive Telematics Software, Verification & Validation, and debug software in a real-time embedded environment with 4G & 5G platforms.
- Analyzed Requirements, created Test Plans, written, and executed Test Cases for various features.
- Experienced in Hardware-in-the-Loop (HIL) setup and Testing. Automated build flashing procedure and knowledge on PKI Injection.
- Supported automation testing needs of new projects, infrastructure upgrades, and application upgrades/enhancements.
- Served as a subject matter expert, acting as a coach and mentor to project teams and technical staff in automation on their assigned project tasks. Experienced with Test Readiness Review (TRR) and test reports.
- Performed manual testing of applications and maintained documentation for testing activities and cases.

- Collaborated with peers to create parameters and comparison data for improved accuracy.
- Read CAN messages through VBE between different ECUs and used CAN analyzer tools to monitor CAN traffic and simulate various scenarios.
- Experienced in installing maps, using the GNSS simulator, to trigger GNSS secondary collision testing.
- Good experience with communication protocols like CAN, LIN, Ethernet, USB.
- Experienced in using GPS Simulator, Cell Simulator (ANRITSU), QXDM logs, and ADB Commands.
- Experienced in reading DLT logs, DLT-Viewer, and applying filters in DLT logs. Automated DLT log capture using Python.
- Experienced using Automotive Hardware like ValueCAN and NeoVI hardware to monitor CAN traffic.
- Performed regression testing and verification of resolved issues. Reported bugs using defect tracking system, JIRA.
- Used test management tools such as RQM to run and gather execution data from automated tests.
- Followed SAFe agile methodology and daily SCRUM planning.
- Developed automated solutions using AWS services such as EC2, S3, Lambda, and RDS to enhance testing processes and data storage capabilities.
- Designed and implemented data transformations using DBT (Data Build Tool) on GCP for improved data quality and maintainability.

Kancy Systems Technology Ltd, Canada-Remote

Software QA Testing Engineer

April 2019 - Dec 2022

Roles and Responsibilities:

- Spearheaded the development of Python test automation frameworks, resulting in the successful automation of test cases and a 30% reduction in manual testing efforts.
- Analyzed business requirements, functional specifications, and other necessary documents to create comprehensive test plans and strategies.
- Leveraged Python and related libraries to write robust test code, ensuring accurate validation of software functionality and adherence to quality standards.

- Conducted OTA (Over-The-Air) testing to ensure the reliability and performance of wireless communication protocols.
- Engineered robust ETL workflows utilizing Python for data extraction from APIs and flat files, performing complex transformations, and loading the refined data into Microsoft SQL Server, enhancing data availability for business intelligence applications.
- Crafted efficient ETL scripts in Python to handle data ingestion, transformation, and storage within MS SQL Server, employing advanced SQL queries and Python data manipulation libraries to maximize processing efficiency and data quality.
- Applied Python and MS SQL Server integration to automate nightly data migration tasks, ensuring timely updates and high availability of data for downstream analytics and machine learning models.
- Collaborated with the development team to identify and resolve issues related to OTA firmware updates.
- Executed data cleansing and manipulation in databases through Pandas, NumPy, and Tableau, ensuring accuracy, with integration of FastAPI, REST APIs, and JSON parsing for smooth data interactions.
- Automated Python test scripts development for GCP services, APIs, and application functionality, prioritizing accuracy and efficiency in verification processes.
- Crafted SQL and PostgreSQL validation tests, guaranteeing data integrity and accuracy in GCP databases, demonstrating a focus on robust data validation techniques.
- Contributed to the improvement of product quality by providing valuable feedback based on OTA test findings.
- Proficiently utilized Appium with Python to conduct comprehensive testing across Android, iOS, and Windows applications, ensuring robust functionality and seamless user experiences.
- Conducted Functional, Negative, Smoke, System, Integration, Regression, UAT, and Performance Testing, applying rigorous testing methodologies to identify and report software defects.
- Collaborated with cross-functional teams to gather feedback and incorporate necessary changes into the automation framework, ensuring continuous improvement and efficient test execution.
- Developed and implemented Behavior-Driven Development

(BDD) frameworks using Python and Selenium WebDriver, effectively cutting down script development time by 50%.

- Utilized Object-Oriented Programming (OOP) concepts and Page Object Model (POM) to create modular and reusable automated test scripts, enhancing test code maintainability and scalability.
- Validated software functionalities by retrieving and manipulating data from Excel, Oracle, and SQL databases, ensuring thorough testing coverage.
- Collaborated with the team to define test data management and environment management approaches, establishing and validating suitable test environments for efficient testing.
- Implemented seamless integration and continuous delivery of software updates using Git and Jenkins, ensuring streamlined deployment processes.
- Created detailed and insightful status reports for senior management, providing them with clear visibility into implementation progress and test coverage.
- Utilized ALM (Application Lifecycle Management) tools to track bugs, meticulously investigating and identifying root causes to facilitate efficient defect resolution.
- Conducted Web service testing using Postman, validating RESTful APIs for functionality and reliability.
- Demonstrated exceptional coordination skills by effectively collaborating with Onsite and Offshore teams, ensuring smooth communication and project success.
- Actively participated in Sprint planning, Sprint Review, Daily Standup, Demo, and Retrospective meetings, contributing to the overall efficiency and progress of the development process.
- Designed and executed API and microservices tests using Python frameworks, like requests, to validate data interactions and integrations within Azure services.
- Integrated automated tests into Azure DevOps pipelines, automating the testing process and providing rapid feedback to development teams.
- Utilized Azure Resource Manager templates to provision and manage test environments for consistent testing conditions.
- Analyzed Requirements, created Test Plans, written, and executed Test Cases for various features.
- Developed and executed test scripts to verify Android-based Infotainment software using Perl and Python-based scripts.

- Developed framework for UI and Image-based verification for Automation scripting in Python.
- Involved in implementing continuous integration and continuous deployment in an automation testing system using Jenkins.
- Experienced with OTA updates and testing.
- Experienced with the Jenkins automation build and test environment.
- Reported bugs using defect tracking system, JIRA. Used test management tools such as Adaptaivist to run and gather execution data from automated tests.
- Reported timely status to the offshore team.
- Coordinated with both onsite and offshore teams to ensure client expectations are well understood and met.
- Followed agile methodology and daily SCRUM planning.
- Developed automated solutions using AWS services such as EC2, S3, Lambda, and RDS to enhance testing processes and data storage capabilities.
- Designed and implemented data transformations using DBT (Data Build Tool) on GCP for improved data quality and maintainability.

VOXCOM LLC, FL- USA

CEO

Aug 2007 - Jan 2019

Roles and Responsibilities:

- Lead Agile methodology implementation, actively participating in QA planning and execution within Sprints.
- Facilitated Scrum meetings and Sprint reviews/retrospectives.
- Led the development and execution of comprehensive test plans and test cases based on requirements from product owners and stakeholders.
- Implemented and supervised Behavior Driven Development (BDD) Testing with Python/Java and Selenium WebDriver to

ensure code reusability and maintainability.

- Directed the development and maintenance of automation frameworks using Selenium WebDriver, Behave, UnitTest, PyTest, and Page Factory for Page Object Model (POM).
- Applied and enforced Object-Oriented Programming (OOP) principles using Python for automation scripting.
- Developed scalable ETL pipelines with Python for batch processing and real-time data ingestion into Microsoft SQL Server, utilizing tools such as Apache Airflow for scheduling and orchestration.
- Implemented data validation and quality checks within Python ETL scripts to ensure the accuracy and reliability of data loaded into MS SQL Server, supporting critical business operations and decision-making processes.
- Leveraged Python's extensive ecosystem, including libraries like PyODBC and pandas, to create ETL solutions that efficiently extract data from various sources, perform necessary transformations, and load it into MS SQL Server databases, optimizing overall data workflow performance.
- Oversaw the implementation and enhancement of Python-based automation scripts.
- In-Vehicle Infotainment Software Expertise:
- Specialized in Automotive In-Vehicle Infotainment Software, leading verification and validation processes.
- Oversaw the analysis of project requirements, creation of detailed test plans, and authoring of test cases for various features.
- Managed the reporting of identified bugs using a defect tracking system, such as JIRA.
- Utilized test management tools, such as Adaptaivist, to run and collect execution data from automated tests.
- Delivered timely status reports to the offshore team.
- Coordinated effectively with both onsite and offshore teams, ensuring a clear understanding of client expectations and successful project delivery.
- Led the team in following agile practices, including daily SCRUM planning, to

maintain an iterative and collaborative approach to software development and testing.

EDUCATION

Independence University - USA

Bachelor Of Science Information Technology

Sep 2015 - May 2019

CERTIFICATIONS

Certification In PCEP

Open EDG- Python Institute